Opgaver Kapitel 13

42. The human eye and the muscles that hold it can be modeled
BIO as a mass–spring system with typical values $m = 7.5$ g and $k = 2.5$ kN/m. What's the resonant frequency of this system? Shaking your head at this frequency blurs vision, as the eyeball undergoes resonant oscillations.

48. A mass is attached to a vertical spring, which then goes into oscillation. At the high point of the oscillation, the spring is in the original unstretched equilibrium position it had before the mass was attached; the low point is 3.0 cm below this. Find the oscillation period.

65. A 340-g mass is mounted on a spring of constant $k = 3.3$ N/m. The damping constant for this system is $b = 8.4\times10^{-3}$ kg/s. How many oscillations will the system undergo before the amplitude decays to $1/e$ of its original value?

70. Two balls with the same unknown mass m are mounted on opposite ends of a 1.5-m-long rod of mass 450 g. The system is suspended from a wire attached to the center of the rod and set into torsional oscillations. If the wire has torsional constant 0.63 N·m/rad and the period of the oscillations is 5.6 s, what's the unknown mass m?

79. You're a structural engineer working on a design for a steel beam,
DATA and you need to know its resonant frequency. The beam's mass is 3320 kg. You test the beam by clamping one end and deflecting the other so it bends, and you determine the associated potential energy. The table below gives the results:

Beam deflection x (cm)	Potential energy U (J)
−4.54	164
−3.49	141
−2.62	71.9
−1.22	9.15
−0.448	0.162
0	0
0.730	4.13
1.29	16.3
2.13	34.0
3.39	115
4.70	225

Find a quantity which, when U is plotted against it, should give a straight line. Make your plot, determine the best-fit line, and use its slope to determine the resonant frequency of the beam.